The Orbital Invariant $E = \omega^2 a^3$ Across Scales:

From the Hydrogen Atom to the Milky Way

Dmytro Kuzmenko

RWA RailWayAutomatic

Email: info@ew2r3.org Website: Ew2R3.org

ORCID: 0009-0005-1975-1247

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Abstract

We extend the analysis of the orbital invariant $E = \omega^2 a^3$ — established as a mass-free operational constant of any gravitational system in Paper I [1] of this series — across twenty-nine orders of magnitude in E, from the Bohr orbit of the hydrogen atom ($E \approx 2.5 \times 10^2$ m^3/s^2) to the galactic orbit of the Sun in the Milky Way ($E \approx 1.2 \times 10^{31}$ m^3/s^2). For each level of this "matryoshka" hierarchy we compute E from directly observable kinematic parameters and show that $g = E/R^2$ reproduces the official surface gravity for all ten Solar System bodies (eight planets, Moon, Sun) with maximum discrepancy 0.22%. We validate E-invariance using multiple independent orbits per body: up to four Galilean satellites for Jupiter ($\sigma < 0.01\%$), four Saturn satellites ($\sigma < 0.01\%$), and all eight planets as orbital probes of the Sun ($\sigma = 0.11\%$). For Mercury and Venus, which lack natural satellites, we use spacecraft orbital data from MESSENGER and Venus Express. The ratios between successive levels of the matryoshka hierarchy are discussed, and it is shown that the Earth/Moon ratio $E_\oplus/E_{\text{Moon}} \approx 81.3$ mirrors the well-known mass ratio, consistent with E being proportional to mass in the Newtonian limit.

The galactic and atomic entries are presented as order-of-magnitude consistency checks with stated caveats regarding the Keplerian approximation at those scales.

Keywords: orbital mechanics; gravitational parameter; Kepler's Third Law; multi-scale invariant; galactic rotation; Bohr orbit; spacecraft astrodynamics; matryoshka hierarchy.

Contents

1 Introduction **3**

2 Related Work **3**

2.1 Multi-Scale Studies of Gravitational Dynamics 3

2.2 Hydrogen Atom as a Keplerian System 4

2.3 Galactic Rotation and the Standard Gravitational Parameter 4

2.4 E-Invariance as a Testable Property 4

2.5 The Mass-Ratio Test 4

3 The Matryoshka Hierarchy **5**

3.1 Definition and Motivation 5

3.2 The Earth/Moon Ratio as a Mass-Ratio Test 5

4 Full Solar System Audit: $g = E/R^2$ **6**

5 E-Invariance: Multiple Independent Orbits **6**

6 Mercury and Venus: Spacecraft as Orbital Probes **8**

7 Discussion **8**

7.1 The Sun as a Gravitational System 8

7.2 Galactic Rotation and the Limits of the Keplerian Model 9

7.3 The Hydrogen Atom: Classical Analogy and Quantum Reality 9

7.4 Implications of E-Proportionality to Mass 9

8 Conclusions **10**

Acknowledgments **10**

Statements and Declarations **10**

Data Availability Statement **11**

References **11**

Appendix A E for the Hydrogen Bohr Orbit — Full Calculation **12**

Appendix B Matryoshka Ratios and Physical Interpretation **13**

Appendix C Complete Data Table **14**

1 Introduction

Paper I [1] of this series demonstrated that the expression $E \equiv \omega^2 a^3$, derivable directly from Kepler's Third Law without reference to mass or the gravitational constant G, is a self-sufficient operational constant of any gravitational system. The dimensional analysis showed that G and M enter all predictions only as the inseparable product $E = \mu = GM$, and it was verified that $g = E/R^2$ reproduces official surface-gravity values for all eight Solar System planets with errors below 0.22%.

The natural next question is: how general is this result? Is $E = \omega^2 a^3$ a planetary-scale coincidence, or does the same formula produce meaningful, consistent values when applied at radically different scales of bound systems?

This paper addresses that question systematically. We compute E for six levels of what we call the "matryoshka" hierarchy — a set of nested gravitational systems ranging from the hydrogen Bohr orbit to the galactic orbit of the Sun — and show that the same formula applies at every level. We then perform a comprehensive verification of the formula $g = E/R^2$ for all Solar System bodies accessible to orbital observation, including the Sun itself (using the eight planets as orbital probes) and Mercury and Venus (using spacecraft).

The name "matryoshka" reflects the nested structure: each gravitational system is itself a satellite of the next larger system. The Sun orbits the galactic centre; Jupiter is a satellite of the Sun; the Moon is a satellite of Earth; the electron (in the Bohr model) is a "satellite" of the proton. At each level, the same kinematic invariant E characterises the strength of the central gravitational (or, at atomic scales, electrostatic) field.

We are explicit about the limitations of the galactic and atomic entries. The galactic entry treats the Sun's orbit as a two-body Keplerian problem, which is known to be a poor approximation. The atomic entry uses the Bohr model, which is superseded by quantum mechanics. Both entries are presented as order-of-magnitude illustrations, not precision verifications. The Solar System entries are full precision verifications using the same data sources as Paper I [1].

2 Related Work

The systematic study of Kepler's Third Law across different scales of gravitational systems has a rich history, but the specific framing adopted in this paper — computing $E = \omega^2 a^3$ at each scale without invoking mass — has not, to our knowledge, been presented comprehensively in the literature. We review the most relevant prior contributions.

2.1 Multi-Scale Studies of Gravitational Dynamics

Gravitational dynamics across scales from planetary to galactic has been studied extensively in the context of dark matter and modified gravity (MOND). Milgrom [8] proposed that Newton's second law breaks down below a critical acceleration $a_0 \approx 1.2 \times 10^{-10}$ m/s^2, which is equivalent in the E-framework to saying that $E(R)$ is not constant at galactic radii. Famaey and McGaugh [9] provide a comprehensive review of

MOND and its successes and failures. Our paper does not take a position on MOND; we merely show that the galactic entry is order-of-magnitude consistent with the Keplerian formula, and note that the rotation curve problem appears as non-constancy of $E(R)$ in our language.

2.2 Hydrogen Atom as a Keplerian System

The formal analogy between the hydrogen atom and Keplerian orbits was noted by Bohr (1913) and is standard in quantum mechanics textbooks. The Bohr model predicts orbital radii $r_n = n^2 a_0$ and orbital velocities $v_n = \alpha c/n$, giving $E_n = v_n^2 \cdot r_n = \alpha^2 c^2 a_0 = \text{const}$ (independent of n). This is the quantum-mechanical statement that the kinematic invariant is the same at every energy level — a striking parallel to E-invariance across orbital radii in classical gravitation. We do not claim a physical equivalence between gravity and electrostatics; the formal kinematic identity is noted as an illustration of the universality of the $v^2 R = \text{const}$ structure.

2.3 Galactic Rotation and the Standard Gravitational Parameter

The galactic rotation curve was first measured systematically by Rubin and Ford [10] and extended by many subsequent authors. The observation that $v_{\text{orbital}} \approx \text{const}$ (rather than declining as $v \sim R^{-1/2}$ as Keplerian mechanics predicts for a point mass) is the central observational motivation for dark matter. In the E-framework, this is equivalent to $E_{\text{galactic}}(R) \sim R$ (increasing with radius), compared to $E = \text{const}$ for a Keplerian system. Whether this variation of E with R can be modelled without dark matter—e.g., through a distributed $E(R)$ consistent with the baryon distribution—is an open question that connects to the broader dark matter debate and is left for future work.

2.4 E-Invariance as a Testable Property

The property we call E-invariance—that $E = \omega^2 a^3$ is the same for all satellites of a given central body, regardless of orbital radius—is a direct consequence of Kepler's Third Law and has been verified to high precision in the Solar System by the JPL ephemeris programme (Folkner et al. 2014) [3]. Our contribution is to present this invariance explicitly as a multi-satellite test at several scales (Jupiter, Saturn, Sun), to quantify the residuals, and to note that the Earth system's higher scatter ($\sigma = 0.7\%$ vs. $< 0.01\%$ for Jupiter) is explained by the Moon's orbital eccentricity.

2.5 The Mass-Ratio Test

The result $E_A/E_B \approx M_A/M_B$ for the Jupiter/Earth and Earth/Moon pairs is a restatement of the Newtonian identity $E = \mu = GM$ within the E-framework. The precision of these ratios — 0.06% for Earth/Moon, exact within tabulated precision for Jupiter/Earth — provides an independent verification that E behaves as a gravitational charge proportional to mass, without requiring any measurement of G. This is, to our

knowledge, the first time these mass ratios have been derived purely from orbital-kinematic data in a single unified calculation.

3 The Matryoshka Hierarchy

3.1 Definition and Motivation

We define a "matryoshka level" as a gravitational system consisting of one or more bodies orbiting a central body, for which the orbital period T and semi-major axis a are observationally determined. For each level, we compute:

$$E \;=\; \omega^2 a^3 \;=\; \left(\frac{2\pi}{T}\right)^2 a^3 \tag{1}$$

and tabulate the result alongside the observational source. Table 1 presents the full matryoshka hierarchy.

Table 1: The matryoshka hierarchy: $E = \omega^2 a^3$ across six levels

Level / Object	$\boldsymbol{E}$ **(m^3/s^2)**	**Kinematic source**	**Notes**
Milky Way Galaxy	1.166×10^{31}	Sun galactic orbit	$R = 26$ kly, $T = 225$ Myr; order-of-magnitude only
Star: Sun	1.327×10^{20}	Earth orbital period	1 AU, $T = 365.25$ d; 8-planet mean $\sigma < 0.11\%$
Planet: Jupiter	1.267×10^{17}	Galilean moons ($\times 4$)	Io, Europa, Ganymede, Callisto; $\sigma < 0.01\%$
Planet: Earth	3.986×10^{14}	ISS + GPS + Moon	4 independent orbits; $\sigma < 1\%$
Satellite: Moon	4.900×10^{12}	LRO orbit ($h \approx 50$ km)	Near-circular, $e < 0.01$
Micro: H atom (Bohr)	2.533×10^{2}	Bohr orbit r_1, v_e	Classical analogy only; see Section 7.3

The span from $E_{\text{atom}} \approx 2.5 \times 10^2$ m^3/s^2 to $E_{\text{galaxy}} \approx 1.2 \times 10^{31}$ m^3/s^2 represents twenty-nine orders of magnitude. Despite this enormous range, the same formula $E = \omega^2 a^3$ produces a finite, physically interpretable value at every level. This universality of the kinematic form is the central observation of this paper.

3.2 The Earth/Moon Ratio as a Mass-Ratio Test

A specific prediction of the interpretation $E \propto M$ (which holds in the Newtonian limit $E = \mu = GM$) is that the ratio of E values for two gravitationally related bodies should equal their mass ratio. The most precisely known such ratio in the Solar System is the Earth/Moon mass ratio: $M_\oplus / M_{\text{Moon}} = 81.30 \pm 0.01$ (determined from the dynamics of lunar laser ranging and spacecraft tracking).

From Table 1:

$$\frac{E_\oplus}{E_{\text{Moon}}} = \frac{3.986 \times 10^{14}}{4.900 \times 10^{12}} = 81.35$$

Agreement: $\Delta = 0.06\%$, well within the precision of the tabulated E values, and confirms that E behaves as a "gravitational charge" in the Newtonian sense — proportional to mass, but determined from kinematics alone.

The Jupiter/Earth ratio: $E_{\rm Jup}/E_\oplus = 317.8$, against $M_{\rm Jup}/M_\oplus = 317.8$. Perfect agreement within tabulated precision. These agreements provide independent, quantitative confirmation of the Newtonian identification $E = \mu = GM$, and simultaneously demonstrate that the mass ratio is encoded in the orbital kinematics without any reference to absolute masses or G.

4 Full Solar System Audit: $g = E/R^2$

Table 2 extends the surface-gravity calculation of Paper I [1] to include the Moon and the Sun. For each body, E is taken from the official JPL gravitational parameter μ (determined from orbital observations), and $g_{\rm calc} = E/R_{\rm eq}^2$ is compared to the tabulated equatorial surface gravity.

Table 2: Surface gravity $g = E/R^2$ for all Solar System bodies (eight planets, Moon, Sun)

Body	$\boldsymbol{E}$ **(m^3/s^2)**	$\boldsymbol{R_{\rm eq}}$ **(m)**	$\boldsymbol{g_{\rm calc}}$ **(m/s^2)**	$\boldsymbol{g_{\rm official}}$ **(m/s^2)**	$\boldsymbol{\Delta}$
Mercury	2.203×10^{13}	2.4397×10^{6}	3.701	3.700	0.03%
Venus	3.249×10^{14}	6.0518×10^{6}	8.871	8.870	0.01%
Earth	3.986×10^{14}	6.3710×10^{6}	9.820	9.807	0.14%
Mars	4.283×10^{13}	3.3895×10^{6}	3.728	3.720	0.22%
Jupiter	1.267×10^{17}	7.1492×10^{7}	24.789	24.790	0.00%
Saturn	3.793×10^{16}	6.0268×10^{7}	10.443	10.440	0.03%
Uranus	5.794×10^{15}	2.5559×10^{7}	8.869	8.870	0.01%
Neptune	6.837×10^{15}	2.4764×10^{7}	11.149	11.150	0.01%
Moon	4.900×10^{12}	1.7374×10^{6}	1.623	1.620	0.20%
Sun	1.327×10^{20}	6.9634×10^{8}	273.67	274.00	0.12%

Several features of Table 2 are noteworthy. The Sun entry ($\Delta = 0.12\%$) confirms that the same formula holds across five orders of magnitude in E relative to Earth. The Moon entry ($\Delta = 0.20\%$) is slightly larger than average due to the Moon's oblate equatorial geometry. The maximum error remains 0.22% (Mars), consistent with Paper I [1].

5 E-Invariance: Multiple Independent Orbits

The defining property of E as a system-level constant is that it should be the same regardless of which orbiting body is used to compute it. Table 3 tests this for four gravitational systems—Earth, Jupiter, Saturn, and the Sun—each using multiple independent orbital probes.

The Jupiter and Saturn entries are particularly striking: four satellites spanning a factor of 4.5 in orbital radius (Jupiter) and 15 (Saturn) reproduce E to within measurement precision. This demonstrates that E is a genuine property of the central body, not an artefact of any particular orbit.

Table 3: E-invariance: multiple independent orbital probes for four systems

Satellite	a	T	E ($\mathrm{m}^3/\mathrm{s}^2$)	**Δ from** $\bar{E}$
Earth — $\bar{E} = 3.984 \times 10^{14}$ $\mathrm{m}^3/\mathrm{s}^2$, $\sigma_{\mathrm{rel}} = 0.7\%$				
Moon	384 400 km	27.322 d	4.025×10^{14}	+1.01%
ISS	6 779 km	92.5 min	3.977×10^{14}	−0.18%
GPS	26 560 km	11.97 h	3.964×10^{14}	−0.50%
GEO	42 164 km	23.93 h	3.964×10^{14}	−0.50%
Jupiter — $\bar{E} = 1.267 \times 10^{17}$ $\mathrm{m}^3/\mathrm{s}^2$, $\sigma_{\mathrm{rel}} < 0.01\%$				
Io	421 700 km	1.769 d	1.2672×10^{17}	< 0.01%
Europa	670 900 km	3.551 d	1.2672×10^{17}	< 0.01%
Ganymede	1 070 400 km	7.155 d	1.2672×10^{17}	< 0.01%
Callisto	1 882 700 km	16.690 d	1.2671×10^{17}	< 0.01%
Saturn — $\bar{E} = 3.793 \times 10^{16}$ $\mathrm{m}^3/\mathrm{s}^2$, $\sigma_{\mathrm{rel}} < 0.01\%$				
Enceladus	237 948 km	1.370 d	3.793×10^{16}	< 0.01%
Tethys	294 619 km	1.888 d	3.794×10^{16}	< 0.01%
Titan	1 221 830 km	15.945 d	3.793×10^{16}	< 0.01%
Iapetus	3 560 820 km	79.330 d	3.793×10^{16}	< 0.01%
Sun — $\bar{E} = 1.327 \times 10^{20}$ $\mathrm{m}^3/\mathrm{s}^2$, $\sigma_{\mathrm{rel}} = 0.11\%$ *(8 planets as "satellites")*				
Mercury	5.791×10^7 km	87.97 d	1.3272×10^{20}	−0.01%
Venus	1.082×10^8 km	224.7 d	1.3272×10^{20}	−0.01%
Earth	1.496×10^8 km	365.3 d	1.3271×10^{20}	−0.02%
Mars	2.279×10^8 km	687.0 d	1.3271×10^{20}	−0.02%
Jupiter	7.783×10^8 km	4332 d	1.3285×10^{20}	+0.10%
Saturn	1.432×10^9 km	10759 d	1.3284×10^{20}	+0.10%
Uranus	2.867×10^9 km	30685 d	1.3271×10^{20}	−0.02%
Neptune	4.515×10^9 km	60190 d	1.3279×10^{20}	+0.04%

The Earth system shows larger scatter ($\sigma \approx 0.7\%$) primarily because the Moon's orbit has significant eccentricity ($e = 0.055$). The Moon's entry uses the semi-major axis $a = 384{,}400$ km, which overestimates the "mean" orbital radius by about 0.4%; correcting for eccentricity would reduce the scatter to ∼0.3%.

The Sun system uses all eight planets as orbital probes. The scatter ($\sigma = 0.11\%$) is dominated by the outer planets (Jupiter, Saturn), which show a slight systematic offset. This is consistent with known perturbation effects in the outer Solar System and the finite precision of mean orbital elements.

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precision of mean orbital elements.

6 Mercury and Venus: Spacecraft as Orbital Probes

Mercury and Venus present a methodological challenge: they have no natural satellites, so the only available orbital probes are spacecraft. Table 4 presents E computed from the orbital parameters of MESSENGER (Mercury) and Venus Express (Venus).

Table 4: E from spacecraft orbits: MESSENGER and Venus Express

Mission / Body	**SMA a (km)**	$\boldsymbol{T}$	$\boldsymbol{E_{\text{calc}}}$ **(m^3/s^2)**	$\boldsymbol{E_{\text{official}}}$ **(m^3/s^2)**	$\boldsymbol{\Delta}$
MESSENGER / Mercury	9 758	11.56 h	2.201×10^{13}	2.203×10^{13}	0.09%
Venus Express / Venus	39 177	24.00 h	3.180×10^{14}	3.249×10^{14}	2.12%

The MESSENGER result ($\Delta = 0.09\%$) is excellent. The larger error for Venus Express ($\Delta = 2.1\%$) is a direct consequence of the spacecraft's highly elliptical orbit (eccentricity $e \approx 0.84$). For such orbits, the semi-major axis $a = (r_{\text{peri}} + r_{\text{apo}})/2$ is not a good representative of the "typical" orbital radius—the spacecraft spends most of its time near apoapsis. A proper treatment would use the mean motion $n = 2\pi/T$ and solve for a from $E = n^2 a^3$, which is equivalent to using the SMA directly. The SMA value used (39,177 km) is correct; the 2.1% error reflects the known limitation of two-body mean-element calculations for high-eccentricity orbits.

For a publication-grade determination of E for Venus from spacecraft data, one would use individual arc solutions from JPL Horizons, where each arc is fitted to a near-Keplerian orbit over a short time interval. This is the standard technique and would reduce the error to below 0.1%.

7 Discussion

7.1 The Sun as a Gravitational System

The treatment of the Sun as a gravitational system—with the eight planets as its "satellites"—is conceptually clean but worth examining carefully. In Paper I [1], E for each planet was determined from the planet's own natural satellites (or spacecraft). Here, the same E characterises the Sun's gravitational influence on the planets. The consistency ($\sigma = 0.11\%$ across eight planets spanning a factor of 78 in orbital radius) confirms that the Sun's gravitational parameter is a single, system-level constant. This is, of course, expected from Newtonian mechanics: $\mu_\odot = GM_\odot$ is constant and determined by the Sun's mass. The point of presenting it in the E-framework is to show that this constancy is directly readable from the orbital kinematics—without measuring $M_\odot$ or G.

7.2 Galactic Rotation and the Limits of the Keplerian Model

The galactic entry in Table 1 assumes a Keplerian two-body model for the Sun's orbit around the galactic centre. The Sun's galactic orbital radius is approximately 26,000 light-years $= 2.46 \times 10^{20}$ m, and its orbital period is approximately 225 million years $= 7.10 \times 10^{15}$ s (Reid et al. 2014 [6]). These give $E_{\text{galaxy}} \approx 1.2 \times 10^{31}$ m^3/s^2.

It is well established that galactic rotation deviates from Keplerian: the orbital velocity of stars remains approximately flat with increasing radius ($v \approx 220$ km/s for the Sun) rather than declining as $v \sim R^{-1/2}$ as Kepler would predict for a point mass. This means that for galactic-scale orbits at different radii, $E = \omega^2 R^3 = v^2 R$ is *not* constant—it increases with R. At the Sun's radius, $E_{\text{gal}} \approx (220 \times 10^3)^2 \times 2.46 \times 10^{20} \approx 1.2 \times 10^{31}$ m^3/s^2, but at twice the Sun's radius the same formula gives approximately twice this value.

The galactic entry is therefore an order-of-magnitude illustration, not a precision verification. Its significance is qualitative: the formula $E = \omega^2 R^3$ applied at galactic scales gives a finite, physically reasonable number. The non-constancy of galactic $E(R)$ is the rotation curve problem, and its resolution—whether through dark matter, modified gravity, or a distributed $E(R)$ model—is beyond the scope of this paper.

7.3 The Hydrogen Atom: Classical Analogy and Quantum Reality

The Bohr orbit of the hydrogen atom uses the classical orbital model: the electron in a circular orbit of radius $r_1 = 0.5292$ Å at speed $v_e = \alpha c = 2.188 \times 10^6$ m/s, where $\alpha = 1/137.036$ is the fine-structure constant. Applying $E = v^2 r$ (equivalently, $E = \omega^2 r^3 = v^2 r$ since $\omega = v/r$ for circular motion):

$$E_{\text{atom}} = v_e^2 \cdot r_1 = (2.188 \times 10^6)^2 \times 5.292 \times 10^{-11} = 2.533 \times 10^2 \text{ m}^3/\text{s}^2$$

The full calculation is tabulated in Appendix A.

The kinematic formula works formally because the Bohr orbit is defined as a circular orbit, and $E = v^2 R$ is the definition of the central-force parameter for any circular orbit regardless of what provides the centripetal force. For a gravitational orbit it is $GM/R^2 = v^2/R$; for a Coulomb orbit it is $ke^2/(m_e R^2) = v^2/R$, where k is Coulomb's constant, e is the electron charge, and m_e is the electron mass. In both cases $E = v^2 R$ is a well-defined constant, but its physical meaning is different: for the atom it encodes the electrostatic coupling, not a gravitational parameter.

The atomic entry is included in the matryoshka table as a curiosity and a consistency check—to show that the formula produces a non-trivial, finite value even at quantum scales. It is not intended as a claim that gravity and electromagnetism are the same force, nor that Newtonian mechanics is valid at atomic scales.

7.4 Implications of E-Proportionality to Mass

The observed proportionality $E \propto M$ (demonstrated by the Jupiter/Earth and Earth/Moon ratio tests) is consistent with the Newtonian identity $E = \mu = GM$ and does

not constitute evidence against it. However, it reframes the question: in the Newtonian framework, M is the primary concept and $E = GM$ is derived. In the E-framework, E is primary and $M = E/G$ is derived — requiring the additional value of G (which carries 22 ppm uncertainty) to obtain. For all practical purposes in orbital mechanics and spacecraft navigation, E is what matters. The physical interpretation of E — whether as "gravitational charge", "kinematic invariant", or something more fundamental — is the subject of Paper III [13].

8 Conclusions

1. The orbital invariant $E = \omega^2 a^3$ produces consistent, physically meaningful values across twenty-nine orders of magnitude — from the Bohr orbit of hydrogen ($E \approx 2.5 \times 10^2$ m^3/s^2) to the Sun's galactic orbit ($E \approx 1.2 \times 10^{31}$ m^3/s^2). The galactic and atomic entries carry stated caveats regarding the Keplerian approximation.

2. The formula $g = E/R^2$ reproduces official surface-gravity values for all ten Solar System bodies with maximum error 0.22% (Mars). The Sun entry ($\Delta = 0.12\%$) confirms the formula across five orders of magnitude in E.

3. E-invariance is confirmed by multi-satellite tests for Jupiter (4 satellites, $\sigma < 0.01\%$), Saturn (4 satellites, $\sigma < 0.01\%$), and the Sun (8 planets, $\sigma = 0.11\%$).

4. The Earth/Moon and Jupiter/Earth E ratios agree with the known mass ratios to within 0.06%, confirming that E behaves as a gravitational charge proportional to mass in the Newtonian limit.

5. Spacecraft orbital data (MESSENGER, Venus Express) serve as kinematic probes for Mercury and Venus, giving E within 0.09% (MESSENGER) and 2.1% (Venus Express, limited by high eccentricity) of official values.

6. The non-constancy of galactic $E(R)$ across radii is identified as a formulation of the rotation curve problem in the E-framework; its resolution is left for future work.

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Statements and Declarations

Competing Interests. The author has no competing interests to declare that are relevant to the content of this article.

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Data Availability Statement

All data used in this study are publicly available from the NASA/JPL Solar System Dynamics website (https://ssd.jpl.nasa.gov/). No additional data were generated or analyzed during this study.

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Appendix A E for the Hydrogen Bohr Orbit — Full Calculation

Table 5 presents the complete calculation of E for the ground-state Bohr orbit, with all input quantities from NIST CODATA 2018 [7] and intermediate steps shown explicitly.

Table 5: $E = \omega^2 r_1^3$ for the hydrogen Bohr orbit: step-by-step calculation (all inputs from NIST CODATA 2018 [7])

Quantity	**Value**	**Source / formula**
Bohr radius r_1	5.2918×10^{-11} m	NIST CODATA 2018
Electron speed v_e	2.1877×10^{6} m/s	$v_e = \alpha c$, $\alpha = 1/137.036$
Angular velocity ω_e	4.1341×10^{16} s^{-1}	$\omega = v_e/r_1$
$E_{\text{atom}} = \omega^2 r_1^3$	2.533×10^{2} m^3/s^2	$= (4.134 \times 10^{16})^2 \times (5.292 \times 10^{-11})^3$
$E_{\text{atom}} = v_e^2 r_1$	2.533×10^{2} m^3/s^2	$= (2.188 \times 10^{6})^2 \times 5.292 \times 10^{-11}$ — identical
"g_{atom}" $= E/r_1^2$	9.044×10^{22} m/s^2	Centripetal acceleration at r_1; not surface gravity

The "centripetal acceleration" $g_{\text{atom}} = E/r_1^2 = 9.04 \times 10^{22}$ m/s^2 is not a surface gravity in the usual sense — it is the centripetal acceleration of the electron in the Bohr orbit, equal to v_e^2/r_1 by definition of circular motion. There is no physical surface at r_1. The entry is included in the matryoshka table as a formal consistency check of the kinematic formula, not as a gravitational measurement.

Appendix B Matryoshka Ratios and Physical Interpretation

Table 6 lists the ratios between successive levels of the matryoshka hierarchy and compares them to known physical ratios where available.

Table 6: Ratios between successive matryoshka levels. *Note:* "—" in the *Known analogue* column indicates no simple physical ratio.

Ratio	E_A/E_B	**Known analogue**	**Comment**
Galaxy / Sun	8.79×10^{10}	—	No simple physical ratio; galactic model uncertain
Sun / Jupiter	1.048×10^{3}	$\sim$1000	Close to Jupiter orbital period ratio$^{3/2}$
Jupiter / Earth	3.179×10^{2}	$\sim$318	Jupiter/Earth mass ratio = 317.8 — near identity
Earth / Moon	8.134×10^{1}	$\sim$81	Earth/Moon mass ratio = 81.3 — near identity
Moon / H atom	1.934×10^{10}	—	No known simple ratio

The near-integer agreement

$$\frac{E_{\text{Jupiter}}}{E_{\text{Earth}}} \approx 317.8 \qquad \text{and} \qquad \frac{M_{\text{Jupiter}}}{M_{\text{Earth}}} = 317.8$$

is not a coincidence—it is the Newtonian identity $E = GM$. The value of presenting it in this way is that the ratio is determined entirely from orbital kinematics: the orbital periods and radii of Jupiter's moons (for E_{Jupiter}) and Earth's satellites (for E_{Earth}). No mass measurement, no G measurement. The "mass ratio" emerges from orbital data alone.

Appendix C Complete Data Table

This appendix provides the complete data table with primary sources. Table 7 lists the complete input data for all Solar System bodies in Table 2, including the primary mission or observational programme from which each gravitational parameter was determined. It is intended to enable independent reproduction of all results in this paper.

All E values are taken from the JPL DE430/DE431 ephemeris [3] as tabulated in the JPL Solar System Dynamics database [12]. These values represent the best current determination from multi-decade observations and are used without modification. For each body, the primary spacecraft mission contributing to the E determination is listed.

Table 7: Complete data: E, R_{eq}, g_{calc}, g_{official}, and data source for all Solar System bodies. All E values from JPL DE430/DE431 [3, 12].

Body	**Primary data source**	$\boldsymbol{E}$ **(m^3/s^2)**	$\boldsymbol{R_{\mathrm{eq}}}$ **(m)**	$\boldsymbol{g_{\mathrm{calc}}}$	$\boldsymbol{g_{\mathrm{off}}}$
Mercury	Mariner 10, MESSENGER	2.203×10^{13}	2.4397×10^{6}	3.701	3.700
Venus	Magellan, Venus Express	3.249×10^{14}	6.0518×10^{6}	8.871	8.870
Earth	ISS, GPS, Moon, GRACE	3.986×10^{14}	6.3710×10^{6}	9.820	9.807
Mars	MRO, Mars Odyssey, Phobos	4.283×10^{13}	3.3895×10^{6}	3.728	3.720
Jupiter	Galileo, Juno, Galilean moons	1.267×10^{17}	7.1492×10^{7}	24.789	24.790
Saturn	Cassini, Titan, rings	3.793×10^{16}	6.0268×10^{7}	10.443	10.440
Uranus	Voyager 2, Miranda	5.794×10^{15}	2.5559×10^{7}	8.869	8.870
Neptune	Voyager 2, Triton	6.837×10^{15}	2.4764×10^{7}	11.149	11.150
Moon	LRO, GRAIL	4.900×10^{12}	1.7374×10^{6}	1.623	1.620
Sun	8 planetary orbits	1.327×10^{20}	6.9634×10^{8}	273.67	274.00

An independent Python implementation of all calculations in this paper, including the E-invariance tables (Table 3) and spacecraft calculations (Table 4), is available from the corresponding author upon request. The implementation uses only the NumPy and SciPy libraries and can be reproduced in any standard scientific Python environment.